CIT3341 E-COMMERCE AND SECURITY

SEMESTER FINAL EXAM

WEDNESDAY, 3RD JUNE 2026

14:00 – 17:00 HOURS

TIME ALLOWED: WRITING – THREE HOURS | READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is compulsory and must be attempted.
2. Section B: Answer THREE (3) questions from this section.
3. This examination paper carries a total of 100 marks.
4. Candidates must not turn this page until the invigilator tells them to do so.

SECTION A – COMPULSORY QUESTION (Total marks for Section A: 40 marks)

INSTRUCTIONS: You MUST attempt ALL questions that follow the case study provided.

"Breached and Exposed: Security Failures in a Growing Online Retail Business"

ShopEasy Zambia, a fast-growing online retail platform, launched operations in 2022 offering consumers the ability to browse products, make purchases, and complete payments entirely online. The platform accepts credit cards, debit cards, and mobile money as payment methods. Within its first year, it had registered over 50,000 customers and processed millions of Kwacha in transactions.

In early 2024, ShopEasy Zambia suffered a significant data breach. Hackers exploited a vulnerability in the company's web application to gain unauthorized access to its customer database. Over 30,000 customer records were stolen, including names, email addresses, phone numbers, and payment card details. The stolen data was later found being sold on an underground online marketplace.

An investigation revealed several security failures. The platform had not implemented **encryption** for stored customer data. There was no **authentication** system beyond basic username and password. The company had also failed to conduct regular vulnerability assessments, and staff were not trained in recognizing phishing attempts. Critically, no **digital certificates** had been deployed to verify the identity of the server to customers.

Customers who discovered the breach reported unauthorized transactions on their payment cards. Several filed complaints with consumer protection authorities, citing violations of their privacy rights. ShopEasy's management admitted they had underestimated the importance of e-commerce security, believing that basic password protection was sufficient to safeguard their platform.

Following the breach, ShopEasy engaged a cybersecurity firm to carry out a full security audit. The firm recommended deploying a Public Key Infrastructure (PKI), implementing Secure Sockets Layer (SSL) certificates, introducing multi-factor authentication, and establishing a formal incident response plan. The firm also emphasized that good e-commerce security requires a balance between strong protection mechanisms and maintaining ease of use for customers.

Requirement:

(a) Identify and explain FOUR core security requirements for e-commerce that ShopEasy Zambia failed to meet, based on the case study above. **(10 marks)**

(b) Explain what encryption is and describe how its absence contributed to the severity of the ShopEasy data breach. Distinguish between symmetric and public key encryption.

(10 marks)

(c) Define a digital certificate and state THREE pieces of information it typically contains.

(10 marks)

(d) List FOUR steps that ShopEasy Zambia should take to improve the security of their platform going forward. **(10 marks)**

(Total: 40 marks)

SECTION B

CHOOSE ANY THREE (03) QUESTIONS FROM THIS SECTION

QUESTION TWO

(a) Define e-commerce and briefly explain how it differs from traditional commerce. Give TWO examples of e-commerce activities. **(4 marks)**

(b) Identify and explain any FOUR of the eight unique features of e-commerce that distinguish it from conventional business. **(8 marks)**

(c) List and briefly describe THREE types of e-commerce business models, giving a real-world example of each. **(6 marks)**

(d) State TWO benefits of e-commerce to consumers and TWO limitations of e-commerce to organizations. **(2 marks)**

(Total: 20 marks)

QUESTION THREE

(a) Define an e-commerce payment system and explain why secure payment systems are essential for online transactions. **(4 marks)**

(b) Distinguish between a credit card and a debit card as methods of online payment. Explain how an online credit card transaction is processed from purchase to settlement.

(6 marks)

(c) Explain what stored value payment systems are and give TWO examples of how they are used in e-commerce. **(4 marks)**

(d) Identify and explain any TWO security requirements that must be met in electronic payment systems to protect both merchants and consumers. **(6 marks)**

(Total: 20 marks)

QUESTION FOUR

(a) Explain what is meant by 'good e-commerce security' and identify the THREE dimensions through which e-commerce security is achieved. **(4 marks)**

(b) Describe the THREE categories of unintentional threats in an e-commerce environment, giving ONE example of each. **(6 marks)**

(c) Identify and explain any FOUR intentional security threats that an e-commerce business may face. **(6 marks)**

(d) Explain the concept of non-repudiation in e-commerce and describe how digital signatures help achieve it. **(4 marks)**

(Total: 20 marks)

QUESTION FIVE

(a) Identify and briefly explain the FOUR major dimensions of ethical, social and political issues that arise in e-commerce. **(4 marks)**

(b) Explain what copyright is and describe TWO ways in which intellectual property rights can be violated in an e-commerce environment. **(6 marks)**

(c) Define a trademark and give TWO examples of how trademarks can be misused in an online business environment. **(4 marks)**

(d) State and briefly explain any THREE social concerns associated with the growth of e-commerce. **(6 marks)**

(Total: 20 marks)

END OF THE EXAMINATION